BRENDA
The Comprehensive Enzyme Information System

Information on EC 3.2.1.8 - endo-1,4-beta-xylanase

TEMPERATURE OPTIMUM	ORGANISM	UNIPROT	COMMENTARY	LITERATURE
100	Bacillus subtilis	-	-	751229
22	Niallia circulans	P09850	assay at	707522
25	Pseudoalteromonas haloplanktis	Q8RJN8	cold-adapted enzyme	656075
30 ⊟	Acetivibrio thermocellus	P10478, O32374	assay at	709200
	Marinimicrobium sp. LS-A18	A0A411LPM1	optimum temperature without NaCl	768777
	Penicillium griseofulvum	-	assay at	707567
35 ⊟	Dickeya chrysanthemi	-	-	655834
	Fusarium graminearum	Q7ZA57	-	663859
	Hordeum vulgare	-	-	171605
	Paenibacillus curdlanolyticus	E3WF08	recombinant enzyme	717395
35 - 45	Talaromyces funiculosus	Q8J0K5	at pH 4.0	664181
37 ⊟	Alternaria sp.	-	assay at	737904
	Arthrinium sp.	-	assay at	737904
	Aspergillus sp.	-	assay at	737904
	Aureobasidium sp.	-	assay at	737904
	Bipolaris sp.	-	assay at	737904
	Cephalosporium sp.	-	assay at	737904
	Chloridium sp.	-	assay at	737904
	Cladosporium sp.	-	assay at	737904
	Cordyceps sp.	-	assay at	737904
	Dothideomycetes sp.	-	assay at	737904
	Epicoccum sp.	-	assay at	737904

	Exophiala sp.	-	assay at	737904
	Fusarium sp.	-	assay at	737904
	Microdochium sp.	-	assay at	737904
	Minimidochium sp. TMS-2011	-	assay at	737904
	Neurospora crassa	-	assay at	737904
	Nigrospora sp.	-	assay at	737904
	Paenibacillus sp. E18	-	assay at	707230
	Paenibacillus sp. W-61	-	assay at	708936
	Penicillium sp.	-	assay at	737904
	Phanerodontia chrysosporium	-	assay at	737904
	Phoma sp.	-	assay at	737904
	Rhodonia placenta	-	assay at	737904
	Sporothrix sp.	-	assay at	737904
	Teredinibacter turnerae	C5BJ89	assay at	749504
	Thermochaetoides thermophila	Q6UN40	-	701091
	Trichoderma reesei	-	assay at	737904
	Trichoderma sp.	-	assay at	737904
	uncultured bacterium	-	-	775836
37 - 40	Euphausia superba	-	isozymes A and B	655257
38	Pseudobutyrivibrio xylanivorans	-	-	655379
40 ⊟	Aspergillus niger	-	assay at	654892
	Bacillus pumilus	-	hydrolysis of larchwood xylan	171624
	Bacillus sp. (in: firmicutes)	-	-	717091
	Bacillus subtilis	P18429	assay at	710466, 709202
	Cellulomonas fimi	-	xylanase B and C	171653

	Cryptococcus sp. (in: basidiomycete fungi)	-	-	171583
	Duganella sp. PAMC 27433	-	-	769515
	Fusarium oxysporum	-	-	171637
		-	assay at	717889
	Hordeum vulgare	Q94G06	-	677960
	Naganishia albida	-	-	171631
	Niallia circulans	P09850	assay at	709203
	Pediococcus acidilactici	-	-	750814
	Penicillium sp. CGMCC 1669	D1GFE6	recombinant enzyme	707207
	Pseudopedobacter saltans	F0S6Y3	-	752084
	Rhizophlyctis rosea	A0A2U8LMQ3	small enzyme form RrXyn11A S	749718
	Streptomyces olivaceus	-	-	746374
	Streptomyces sp.	F7J663	-	730370
	Triticum aestivum	Q9XGT8	-	752127
	uncultured bacterium	A0A3G9FT71	-	750245
42	Aspergillus niger	-	-	171599
45 ⊟	Albifimbria verrucaria	-	-	171652
	Arthrobacter sp.	A0A0F6MV31	-	737475
	Aspergillus luchuensis	P33557	-	739002
	Aspergillus niger	-	-	171598
		-	xylanase I and II	171597
		I3QKR9	wild-type enzyme	751243
	Bacillus subtilis	-	at pH 6.0	770731
	Carica papaya	Q8GTJ2	about	655665
	Cellulomonas fimi	-	xylanase A	171653
	Luteimicrobium xylanilyticum	A0A286LNP2	C-terminal truncated mutant	745089
	Malbranchea pulchella	-	xylanase A and B	171587

	Massilia sp. RBM26	A0A0S2I9J6	-	738866
	Neocallimastix patriciarum	A8TGA1	-	695791
	Paenibacillus terrae	F1KBQ3	enzyme without xylan-binding domain	750448
	Penicillium occitanis (nom. inval.)	-	-	718338
	Pomacea insularus	-	-	171584
	Pomacea maculata	Q7Z1V6	recombinant EGXA with xylan as substrate	695295
	Sphingobacterium sp. TN19	D8L2Y2	-	707250
	Trichoderma longibrachiatum	-	-	171647
	Trichoderma virens	-	-	171632
45 - 50	Bacillus pumilus	-	-	171586
50 ⊟	Acetivibrio thermocellus	-	-	750432
	Alternaria mali	-	-	695683
	Aspergillus fumigatus	-	-	737935
		E0X4B3	wild-type and mutants S68R and Q47P/S159R	738888
	Aspergillus giganteus	-	both xylanase I and xylanase II	665016
	Aspergillus niger	-	-	171634
		-	xylanase II	171633
		-	-	664489, 707403, 731024
		F5CI28	-	729014
	Aspergillus ochraceus	-	-	171654
	Aspergillus sulphureus	Q3S401	recombinant enzyme	707197
	Aureobasidium melanogenum	-	-	776226
	Bacillus licheniformis	-	-	752334
	Bacillus pumilus	Q5EFR9	recombinant enzyme	717410

Bacillus sp. (in: firmicutes)	-	-	667012, 708851
Bacillus subtilis	-	-	171618, 171622
Cellulomonas flavigena	A1XM14	-	697503
	A1XM14	xylanase activity	697503
Cellvibrio mixtus	O68541	-	744626
Cladosporium sp.	A0A346PXS9	-	750679
Cohnella sp. AR92	-	-	768810
Cutaneotrichosporon cutaneum	-	-	171625
Kitasatospora sp.	A0A3T0YV67, A0A3T0YV60	-	750084
Lentinula edodes	-	Xylam and Xyl6-6, pH 5.0	655228
	-	-	681351
Marinimicrobium sp. LS-A18	A0A411LPM1	optimum temperature with 10% (w/v) NaCl	768777
Neocallimastix patriciarum	P29127	-	744626
Paenibacillus curdlanolyticus	A0A0A0R2Z1	-	737855
Paenibacillus sp.	B6VF01	-	717404
Penicillium citrinum	-	at both pH 3.5 and pH 8.5	681711
	B1B533	-	699042
Penicillium janczewskii	-	-	739567
Penicillium sclerotiorum	-	xylanase I	707211
Rhizophlyctis rosea	A0A2U8LMQ3	large enzyme form RrXyn11A L	749718
Ruminiclostridium josui	-	substrate: alpha-D-cellobiose octaacetate. Enzyme derivative RjAbf62A-CBM6, containing the glycoside hydrolase family 62 (GH62) catalytic module and the family 6	750435

			carbohydrate binding module (CBM6)	
	Ruminococcus flavefaciens	-	catalytic domains XYL-A and XYL-C	171648
	Schizophyllum commune	-	-	171590, 664785
		P35809	-	729654
	Streptomyces sp.	F8WSY7	-	729010
	Termitomyces clypeatus	-	-	707210
	Thermoascus aurantiacus	-	assay at	656559
	Thermomyces lanuginosus	O43097	carboxymethyl cellulase activity of the fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus and of the parental endoglucanase	750783
	Thermotoga maritima	-	assay at	656575
	Trametes hirsuta	-	30 min, stable	171629
	Trichoderma harzianum	-	20000 Da protein	171613
	Trichoderma inhamatum	-	-	695668
	Trichoderma reesei	-	assay at	656334
		P36218	assay at	657221
		B2CNY5	-	709576
	uncultured bacterium	-	-	737477
		A0A127JBT0	-	744542
50 - 55 ⊟	Aspergillus caespitosus	-	both isoforms xyl I, xyl II	666701
	Bacillus subtilis	-	-	709497
	Cellulomonas flavigena	-	wild-type enzyme	655370
50 - 60 ⊟	Acidomyces acidophilus	-	wild-type and mutant enzymes	707765
	Aspergillus niger	-	optimum temperature for hybrid xylanase and	695678

			its Aspergillus parent is 60°C and 50°C, respectively	
	Penicillium griseofulvum	-	-	677745
	Talaromyces funiculosus	Q9HFH0	-	677745
	Thermobifida fusca	Q56265	hybrid gene from Thermomonospora fusca with N-terminal region of Bacillus subtilis, temperature optimum is similar to temperature optimum of Thermomonospora fusca parent enzyme	664701
55 ⊟	Acrophialophora nainiana	-	-	656332
	Alicyclobacillus sp.	D3VWB5	-	717947
	Aspergillus cf. niger BCC14405	Q6QA21	-	665441
	Aspergillus flavus	B8NKE9	-	752113
		B8NJ86	-	750988
	Aspergillus niger	-	-	679573
	Aspergillus versicolor	-	-	666703
	Bacillus pumilus	-	-	769914
	Bacillus sp. (in: firmicutes)	-	recombinant enzyme from	171650
	Bacillus sp. 41M-1	Q9RC94	wild-type DELTAXBD	696858
	Bacillus subtilis	-	xylanase XynA	719992
	bacterium enrichment culture clone Xyl8B8	H9ZGD1	-	730872
	Cellulomonas flavigena	-	mutant enzyme	655370
		-	-	746350
	Cellulosimicrobium cellulans	T2M1V1	-	731029
	Demequina sp. JK4	B9VSZ3	-	709491
	Fusarium oxysporum	-	-	663671
	Fusarium proliferatum	-	-	657201

Geobacillus stearothermophilus	-	-	680969
Kitasatospora sp.	A0A3T0YV67, A0A3T0YV60	-	750084
Kocuria sp. Mn22	B8XY24	-	709503
Melanocarpus albomyces	-	isozyme IIc	654986
Neocallimastix patriciarum	Q9UV68	wild-type	744133
Nesterenkonia xinjiangensis	D1KJJ7	recombinant enzyme	707216
Niallia circulans	P09850	-	769727
Paenibacillus terrae	F1KBQ3	native enzyme	750448
Penicillium citrinum	Q2PGY1	-	665763
	Q2PGY1	recombinant enzyme	752115
Penicillium sclerotiorum	-	xylanase II	707211
Pleurotus ostreatus	A0A2U9Q8K4	-	750334
Ruminiclostridium josui	-	RjAbf62A-Axe6A, containing all four modules but lacking the signal peptide	750435
Streptomyces sp.	-	-	171619
Streptomyces thermocarboxydus	C6ZHB0	recombinant enzyme	709555
Talaromyces pinophilus	-	all 3 isozymes	655807
	A0A6N4SL16	-	768726
Termitomyces clypeatus	-	-	171630
Thermobifida fusca	-	-	707403
Thermochaetoides thermophila	G0S9G3	-	750799
	G0S9G3	for CTendo7's xylanase activity and endoglucanase activity	750799
Thermothelomyces thermophilus	-	-	742765
Trichoderma reesei	LC132960	wild-type	744124

	Trichoderma sp.	-	isozyme EX2	707762
	Trichoderma sp. TP3-36	-	-	776215
55 - 60 ⊟	Acrophialophora nainiana	-	-	655620
	Mycothermus thermophilus	-	xylanase 1, xylanase 2	738127
	Streptomyces sp.	-	-	171660
55 - 62	Trichoderma reesei	-	-	680210
56	Aspergillus nidulans	-	hydrolysis of oat spelt xylan	171658
58	Cellulomonas uda	-	-	136300
60 ⊟	Alicyclobacillus sp.	A0A0P0ISK8	-	737905
	Aspergillus fumigatus	E0X4B3	mutant A60D	738888
		-	-	769929
	Aspergillus japonicus	-	-	709190
	Aspergillus niger	-	-	710686
		P55329	-	744626
		I3QKR9	mutant enzyme T205C/A52C	751243
	Bacillus cereus	-	the enzyme activity at 50°C is slightly lower than at 60°C, but no significant difference between 50 and 60°C	770012
	Bacillus licheniformis	-	-	682685
	Bacillus sp. (in: firmicutes)	-	-	171656
		Q84F19	-	664690
	Bacillus subtilis	Q45VU2	assay at	707810
		A0A172MAU1	immobilized enzyme and soluble enzyme	750593
	Bispora sp. MEY-1	D6MYS9	-	707258
	Cellulosimicrobium sp.	B2BZ80	-	737466
	Evansstolkia leycettana	-	-	774449

Gelatoporia subvermispora	-	-	664998
Halalkalibacterium halodurans	-	-	681342
Irpex lacteus	-	xylanase I	171612
Lentinula edodes	-	Xyl1-3, pH 5.0	655228
Melanocarpus albomyces	-	isozymes IIa and IIb	654986
Microbacterium trichothecenolyticum	A0A067XRW0	-	730191
Paenibacillus campinasensis	Q2I6W5	-	718337
Paenibacillus curdlanolyticus	B1A3N2	-	699498
Paenibacillus sp.	-	-	717094
Phanerodontia chrysosporium	-	isoform xynB	664683
Pseudozyma hubeiensis	-	assay at	707766
Rhodothermaceae bacterium RA	A0A1V0DJ74	-	775585
Streptomyces avermitilis	Q9X584	purified heavier protein band (42.8 kDa)	698373
Streptomyces coelicolor	Q8CJQ1	-	737481
Streptomyces fradiae	A7TVD4	at pH 7.8	699489
Streptomyces griseorubens	-	-	745416
Streptomyces halstedii	-	-	171639
Streptomyces lividans	-	-	171606, 171608
Streptomyces olivaceoviridis	-	-	171659
Streptomyces sp.	-	-	654144, 655355
Streptomyces sp. S9	B4XVN1	-	695801
Streptomyces sp. T7	-	-	774465
Streptomyces sp. TN119	C6FX34	recombinant enzyme	707196

	Streptomyces thermocyaneoviolaceus	Q9RMM4	native enzyme	707809
	synthetic construct	-	-	696797
	Talaromyces purpureogenus	A0A2H4RAC5	recombinant enzyme	749635
	Thermoascus aurantiacus var. levisporus	-	-	743330
	Thermobifida fusca	-	hybrid xylanase and its Thermomonospora parent	695678
	Thermochaetoides thermophila	-	xylanase II	737964
	Thermomyces lanuginosus	-	wild-type enzyme	655088
		-	-	696797
	Thermothelomyces thermophilus	G2QIK8, G2QDB9	recombinant enzyme	750646
	Trichoderma harzianum	-	29000 Da protein	171613
	Trichoderma reesei	B2CNY5	-	707864
		-	Xyn2	707911
		-	recombinant wild-type enzyme	710126
		LC132960	mutant Q286A/N340Y	744124
	uncultured microorganism	I6M1A2	-	731026
60 - 65 ⊟	Streptomyces cyaneus	-	-	678757
	Streptomyces sp.	-	-	171592, 171594
	Trichoderma harzianum	-	20000 Da enzyme form	171613
65 ⊟	Aspergillus terreus	-	-	707209
	Aspergillus tubingensis	-	-	707209
	Bacillus sp. (in: firmicutes)	-	enzyme component I from strain W1 and strain W2	171591, 171595

	-	enzyme from strain W1 and W3	171621
Bacillus subtilis	-	chimeric enzyme CorAXynA	719992
Luteimicrobium xylanilyticum	A0A286LNP2	wild-type	745089
Melanocarpus albomyces	-	isozyme IId	654986
	-	-	738730
	-	both isoforms xylanase IA and xylanase IIIA	738730
Melioribacter roseus	I7A603, I6Z823, I6Z9A7	-	738862
Microcella alkaliphila	A0A0U5B5Z4	-	738192
Mycothermus thermophilus	-	-	743097
Neocallimastix patriciarum	Q9UV68	mutant N38Y/F52W/G56Y/G201L	744133
Paecilomyces variotii	-	-	738999
Paenibacillus sp.	A0A075EEX4	-	730187
Remersonia thermophila	-	-	729007
Streptomyces olivaceoviridis	-	-	707403
Streptomyces sp. S27	-	-	695696
Thermoascus aurantiacus	-	wild-type and mutants Q158A, H209N	738405
Thermomyces lanuginosus	-	-	171617, 737547
	O43097	wild-type	742096
	-	-	742958
	-	wild-type	743887
	O43097	parental xylanase	750783
Thermotoga maritima	-	recombinant hybrid mutant	710126
Trametes hirsuta	-	-	171629

	Trichoderma reesei	-	recombinant hybrid mutant	710126
		-	recombinant enzyme	707415
	uncultured bacterium	B2LWN3	native enzyme	698785
	uncultured microorganism	E2DQY5	-	717134
65 - 80	Aspergillus niger	-	xylanase IA and IB	171633
70 ⊟	Acetivibrio thermocellus	-	-	738886
	Acidothermus cellulolyticus	A0LR95	-	750497
		A0LR95	recombinant enzyme	750497
	Aspergillus fumigatus	E0X4B3	mutant T16A/T39I/L176Q	738888
		-	-	742216
	Bacillus licheniformis	-	X-I	677623
	Bacillus sp. (in: firmicutes)	-	-	171595, 171650
		-	enzyme component II from strain W1 and strain W2	171591
		-	enzyme form strain W2 and W4	171621
		-	-	654996, 663532, 682728, 717105
	Bacillus subtilis	-	thermostable variant XynAG3 and chimeric enzyme CorAXynAG3	719992
	Caldicellulosiruptor saccharolyticus	-	-	171655
	Halalkalibacterium halodurans	-	at pH 10	679547
		-	-	681342, 731022
	Irpex lacteus	-	xylanase III	171612
	Malbranchea cinnamomea	-	-	741623

Malbranchea pulchella	-	-	171587, 171596
Melanocarpus albomyces	-	isozymes Ia, Ib, and Ic	654986
Penicillium canescens	C3VEV9	-	729182
Phanerodontia chrysosporium	-	isoforms xynA, xynC	664683
Phialophora sp. CGMCC 3328	F2VRY7	-	717131
Rasamsonia byssochlamydoides	-	xylanase X-b-I and X-b-II	171627
	-	xylanase bI	738999
	-	xylanase bII	738999
Saccharopolyspora pathumthaniensis	E1APH5	-	717401
Streptomyces halstedii	Q59922	-	751704
Termitomyces aurantiacus	-	-	742400
Thermoanaerobacterium saccharolyticum	P36917	-	775899
Thermoascus aurantiacus	-	mutant N257D	738405
Thermochaetoides thermophila	Q8J1V6	-	709894
	-	xylanase I	737964
Thermomyces lanuginosus	-	mutant enzyme	655088
	-	-	742766, 742165, 742400
	-	mutant M7	743887
	O43097	xylanase activity of the fusion enzyme (EG-M-Xyn) of endoglucanase (cellulase) from Teleogryllus emma and xylanase from Thermomyces lanuginosus	750783

	Thermopolyspora flexuosa	-	recombinant His-tagged enzyme	707229
	Thermothelomyces fergusii	-	-	741623
	Thermothelomyces heterothallicus	-	-	655176
	Thermothelomyces thermophilus	-	-	742216
	Trichoderma reesei	-	at pH 4.0	770736
	Trichoderma sp.	-	isozyme EX1	707762
	[Caldibacillus] cellulovorans	Q9L8L8	recombinant catalytic domain	656718
70 - 75	Thermomyces lanuginosus	-	-	171643
75 ⊟	Acetivibrio thermocellus	A3DJS9	-	737360
	Bacillus licheniformis	-	X-II	677623
	Bacillus sp. (in: firmicutes)	-	-	171645
	Geobacillus stearothermophilus	-	-	171651
	Halalkalibacterium halodurans	-	at pH 9	679547
		-	-	774238
	Paecilomyces sp. 'thermophila'	D1G4K3	recombinant His-tagged enzyme	707603
	Rasamsonia byssochlamydoides	-	xylanase X-a	171627
		-	xylanase A	738999
	Rhizomucor miehei	-	-	717022
	Streptomyces sp.	D1FNQ6	-	746234
	Thermoascus aurantiacus	-	-	742421
	Thermoclostridium stercorarium	-	-	171615
		-	xylanase A	171582

	Thermomyces dupontii	-	-	741629, 741630
	Thermomyces lanuginosus	O43097	native and recombinant enzyme	654274
		O43097	mutant Q1C/Q24C	742096
		-	-	646829
75 - 80 ⊟	Streptomyces ipomoeae	MK331807	-	750086
	Thermomyces dupontii	-	-	742661
76	Thermoascus aurantiacus	-	-	741753
80 ⊟	Actinomadura sp.	-	recombinant XynS14	717886
	Bacillus amyloliquefaciens	-	-	171646
	Bacillus sp. (in: firmicutes)	-	-	171628
	Halalkalibacterium halodurans	M4QNR9	-	737851
	Laetiporus sulphureus	-	-	699053
	Malbranchea cinnamomea	A0A088BC14	-	742680
	Melioribacter roseus	I7A603, I6Z823, I6Z9A7	-	738862
	Parageobacillus thermantarcticus	-	-	657331
	Pseudothermotoga thermarum	F7YXD6	-	729475
	Talaromyces funiculosus	Q5ZNB1	-	718080
	Thermoascus aurantiacus	-	-	171601, 742959
	Thermobifida alba	-	-	171636
	Thermobifida fusca	Q47QL8	recombinant protein, both with and without His-tag	737932
	Thermoclostridium stercorarium	-	-	171657

		-	hydrolysis of xylan	171661
	Thermoclostridium stercorarium subsp. thermolacticum	-	-	171649
	Thermomonospora sp.	-	-	654413
85 ⊟	Bispora sp. MEY-1	D0QF43	recombinant Xyl10C	708455
	Priestia megaterium	-	-	681810
	Thermothielavioides terrestris	-	-	742470
90 ⊟	Acidothermus cellulolyticus	-	-	717123
	Aspergillus fumigatus	-	-	682991
	Caldicellulosiruptor owensensis	E4Q2A4	-	745147
	Halalkalibacterium halodurans	-	-	729408
	Lentinula edodes	-	Xyl16-3, pH 5.0	655228
	Saccharolobus solfataricus	-	-	679721
		Q97YG7	-	692258
	Thermotoga maritima	-	-	663571
	Thermotoga petrophila	A5IL00	-	746962
	[Caldibacillus] cellulovorans	Q9L8L8	recombinant N-terminal + catalytic domains	656718
95	Thermotoga petrophila	A5IL00	-	748630
additional information	Pseudoalteromonas haloplanktis	Q8RJN8	psychrophilic enzyme with high activity at low temperatures and low thermal stability	656075